



NITI Aayog



SECTOR AT A GLANCE 02

Solar Energy in India

Energy in India

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SECTOR AT A GLANCE 02

SOLAR ENERGY IN INDIA

ENERGY IN INDIA

The **'Sector-at-a-Glance'** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector

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Overview

In the past five years, India has made remarkable progress in solar energy, becoming a global leader. Solar energy constitutes the largest portion (54%) of India's total renewable energy capacity as of 2023 and India ranks fourth worldwide in photovoltaic (PV) deployment, with its solar capacity growing by around 200%, from about 21.5 GW in 2018 to approximately 64.3 GW in March 2023.¹

Scaling efforts in solar energy adoption can accelerate India's energy security journey, meet energy needs in remote and rural areas, support the growth of sectors like green hydrogen and electric mobility, and contribute to the green jobs market. Despite progress, challenges persist with the pace of solar capacity installation falling behind the required rate. From 2018 to 2023, annual solar installation has increased from ~5GW to ~13GW, yet it remains half of the annual target (30 GW) needed to achieve the 2030 goal of 270 GW.² Additionally, domestically manufactured PV modules remain more expensive than their Chinese counterparts, impacting the competitiveness of domestic manufacturers.³

Given these opportunities and challenges, domestic solar component manufacturing and promoting solar power adoption across sectors are the major policy focus areas. Several ongoing and recent central government initiatives, including but not limited to production-linked incentives and PM-KUSUM focus on incentivising domestic manufacturing and incentivising use of solar agricultural pumps. These efforts along with several state- and local-level policies and initiatives, seek to establish India as a global leader in solar energy.

Promoting technological and workforce development, enhancing transmission and storage infrastructure across states to enable effective grid integration, and promoting rooftop solar adoption will be critical to continue advancing India's solar ambitions. This note presents key trends related to different components of solar energy in India and actions for the way forward.

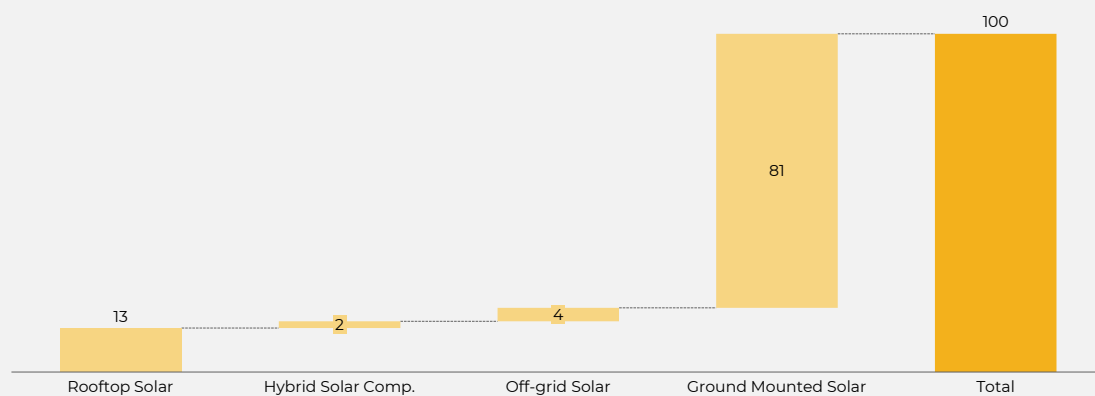
Figure 1: Key solar energy statistics



Source: Ministry of New & Renewable Energy, Physical Progress, 2023; Ministry of New and Renewable Energy, Solar Energy capacity has nearly tripled in last 5 years from 21651 MW to 64380 MW, 2023; Ministry of New and Renewable Energy, State-wise RE installed capacity, accessed on November 2023; Ministry of New and Renewable Energy, Top Policymakers to speak on "India PV EDGE 2020" on Solar Manufacturing, 2020

¹ Ministry of New & Renewable Energy, Physical Progress, 2023; Ministry of New & Renewable Energy, Solar Energy, accessed July 2023; Ministry of New and Renewable Energy, Solar Energy capacity has nearly tripled in last 5 years from 21651 MW to 64380 MW, 2023
² Ministry of New and Renewable Energy, Year wise achievements, accessed December 2023; Ministry of New and Renewable Energy, Solar Energy capacity has nearly tripled in last 5 years from 21651 MW to 64380 MW, 2023
³ CEEW, Scaling Up Solar Manufacturing to Enhance India's Energy Security, 2020

Figure 2: Installed solar capacity as a percentage share of the total by type of solar power-producing entity (2023)



Source: Ministry of New and Renewable Energy, State-wise RE installed capacity, accessed November 2023

Key trends

This section dives deeper into India's solar energy landscape, outlining statistics and trends on:

1. Resourcing, planning, and data management
2. Manufacturing, assembly, and standards
3. Generation, integration, and storage
4. Solar power consumption
5. Workforce development
6. Waste management
7. Research and development
8. International collaboration
9. Monitoring and evaluation

Resourcing, planning, and data management

Experiencing massive growth in solar energy over the past decade, India has set ambitious 2030 targets.

India's solar potential is about 750 GW⁴:

India's estimated solar power generation potential stands at about 750 GW, i.e., ~50% of the total renewable power generation potential as of March 2022.⁵ This estimate excludes rooftop solar.

Indian states actively collaborate for better renewable energy planning:

State-nodal agencies in India have accumulated significant expertise and know-how in the planning and execution of renewable energy (RE) initiatives. To facilitate mutual learning and the exchange of best practices, technological insights, and program knowledge around solar energy, the Ministry of New and Renewable Energy (MNRE) established the Association of Renewable Energy Agencies of States (AREAS). The association includes three Standing Committees: Technology and Resource Assessment, Policy and Finance, and IT & IMS.

Solar comprises more than half of the total installed renewable capacity:

Solar power comprises 53.5% of renewable energy capacity, as of 2023.⁶ Solar energy capacity in India witnessed astounding growth over the past seven years, displaying a 2,447% increase from 2.63 GW in March 2014 to ~67 GW as of May 2023.⁷

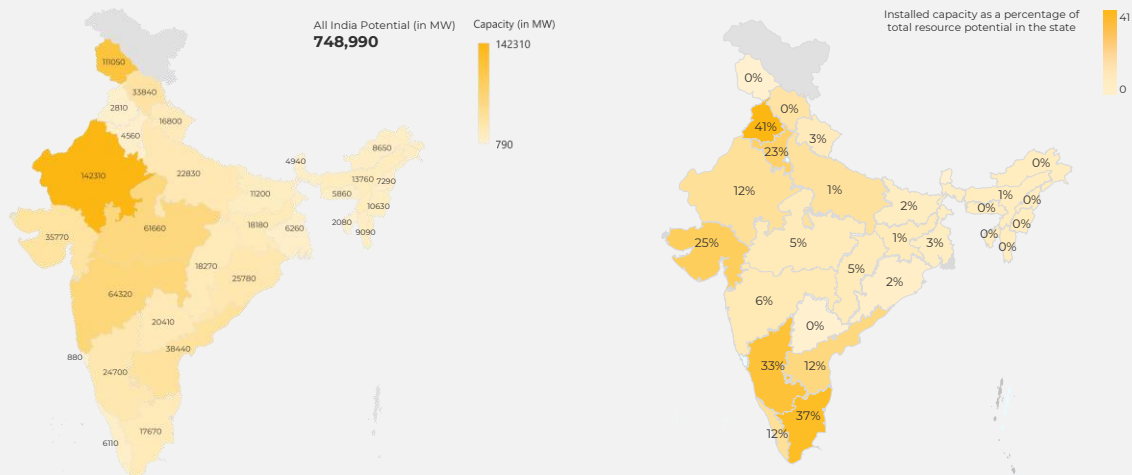
⁴ Ministry of Information and Broadcasting, Union Budget 2022-2023: India Embarks on a Solar Journey, 2022

⁵ Ministry of Statistics and Programme Implementation, Energy Statistics India- 2023, 2023; Vice President's Secretariat, Vice President calls for increasing the production of Solar PV cells and modules in India, 2021

⁶ Ministry of New & Renewable Energy, Physical Progress, 2023

⁷ Ministry of Information and Broadcasting, Union Budget 2022-2023: India Embarks on a Solar Journey, 2022; Ministry of New & Renewable Energy, Physical Progress, 2023

Figure 3: Solar power potential and installed capacity as a share of total resource potential by state and UT (2023)



Source: NITI Aayog, Indian Climate and Energy Dashboard, accessed December 2023

Manufacturing, assembly, and standards

Domestic solar component manufacturing faces challenges competing with mature and advanced foreign competitors. However, recent policy efforts have helped promote a rapidly growing domestic manufacturing ecosystem.

India's domestic PV manufacturing potential has more than doubled between 2020 and 2023:

India's solar module manufacturing capacity is expected to grow fourfold by 2025 compared to 2021.⁸ By 2026, India is projected to reach a manufacturing capacity of 110 GW, achieving self-sufficiency in meeting the demand for PV modules.⁹ Leading players in the PV manufacturing industry include Waaree, Adani Solar, and Vikram Solar.¹⁰

There is a robust enabling environment for solar production in India:

The government's Production Linked Incentive (PLI) scheme has catalysed a robust solar module manufacturing ecosystem in India, aiming to reduce import dependence. The scheme is estimated to add 65,000 MW of PV manufacturing capacity annually. It is also expected to bring a direct investment of ~94,000 crores and create direct employment of 1,95,000 people.¹¹

Established standards have driven quality compliance and promoted global-standard domestic manufacturing:

MNRE issues orders detailing quality standards, and the Bureau of Indian Standards (BIS) takes on implementation through testing products in BIS-recognised labs. These products, like solar PV modules and battery storage devices can be used in solar projects only if registered with BIS.¹²

⁸ Ministry of Information and Broadcasting, Union Budget 2022-2023: India Embarks on a Solar Journey, 2022

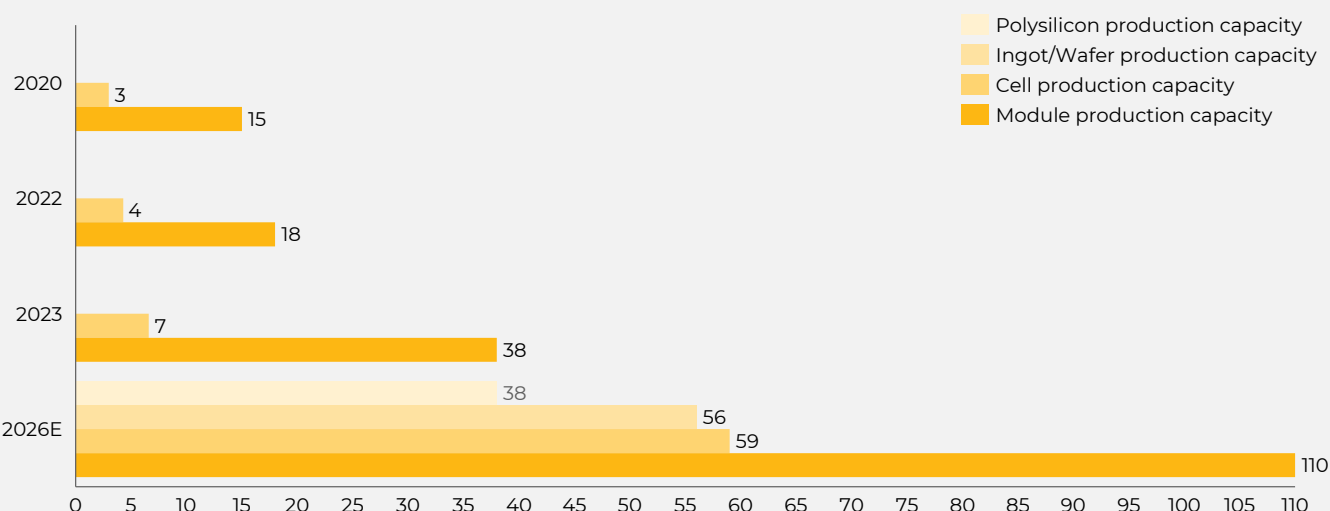
⁹ IEEFA, India's photovoltaic manufacturing capacity set to surge, 2023

¹⁰ JMK and IEEFA, India's Photovoltaic Manufacturing Capacity Set to Surge, 2023

¹¹ Cabinet, Cabinet approves Production Linked Incentive Scheme on 'National programme on High Efficiency Solar PV Modules' for achieving manufacturing capacity of Giga Watt (GW) scale in High Efficiency Solar PV Modules, 2022; Ministry of Power, Government allocates 39600 MW of domestic Solar PV module manufacturing capacity under PLI, 2023

¹² Ministry of New and Renewable Energy, Annual Report 2021-22, 2021

Figure 4: Growth of domestic PV manufacturing capacity in India in GW (2023)



Source: IEEFA, India's Photovoltaic Manufacturing Capacity Set to Surge, 2023

Generation, integration, and storage

With the rising share of solar in the electricity mix, India will face issues of integrating large-scale variable RE into the grid.

In 2023-24, solar power projects generated approximately 116 billion kWh of electricity:

In India, power generation from solar energy increased by 131% from 2019-20 to 2023-24¹³ By 2027, solar energy is projected to contribute to approximately 9% of the overall electricity generation (utility) and is expected to increase to around 19% by 2047.¹⁴

The central and state governments have established several measures to promote solar power generation and support grid integration:

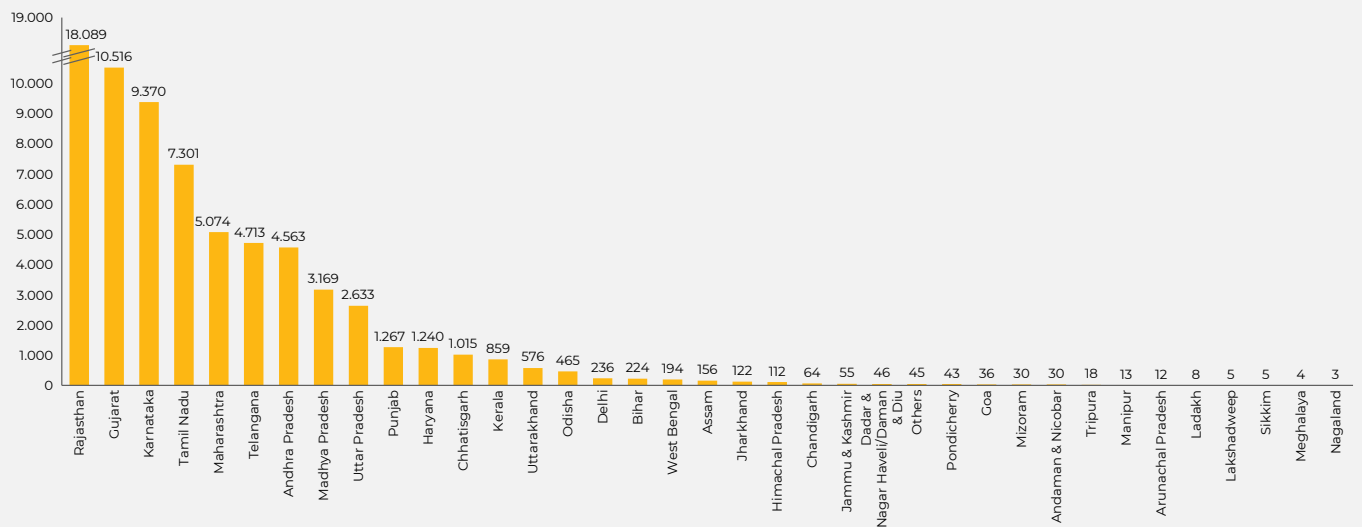
These include a) The Solar Parks Scheme, which targets 40 GW of installed capacity by March 2024, b) Waiver of Inter State Transmission System (ISTS) charges for inter-State sale of solar and wind power for projects to be commissioned by 30th June 2025, c) Foreign Direct Investment (FDI) under automatic route in solar power generation, d) Notification of standards for deployment of solar photovoltaic system/devices, e).Setting up of project development cell for attracting and facilitating investments, f) Off-grid Decentralized and Solar PV Applications Programme, g) Floating solar projects. Additionally, the MNRE has also implemented two significant schemes for decentralised solar power generation. The PM KUSUM scheme aims to deploy 10,000 MW of capacity through grid-connected solar power plants and pumps. Additionally, the Green Energy Corridor –launched in 2015 and now in its second phase – supports grid integration and power evacuation for 100 GW of solar energy across 21 states.¹⁵

¹³ Central Electricity Authority, Monthly Renewable Energy Generation Report, 2024; Ministry of New and Renewable Energy, BU Solar Energy produced this financial year till January, 2023; SUPRABHA and NISE, National Online Rooftop Solar Data Monitoring Centre (NORS-DMC), 2019

¹⁴ NITI Aayog, India Energy Security Scenarios 2047, Accessed August 2023

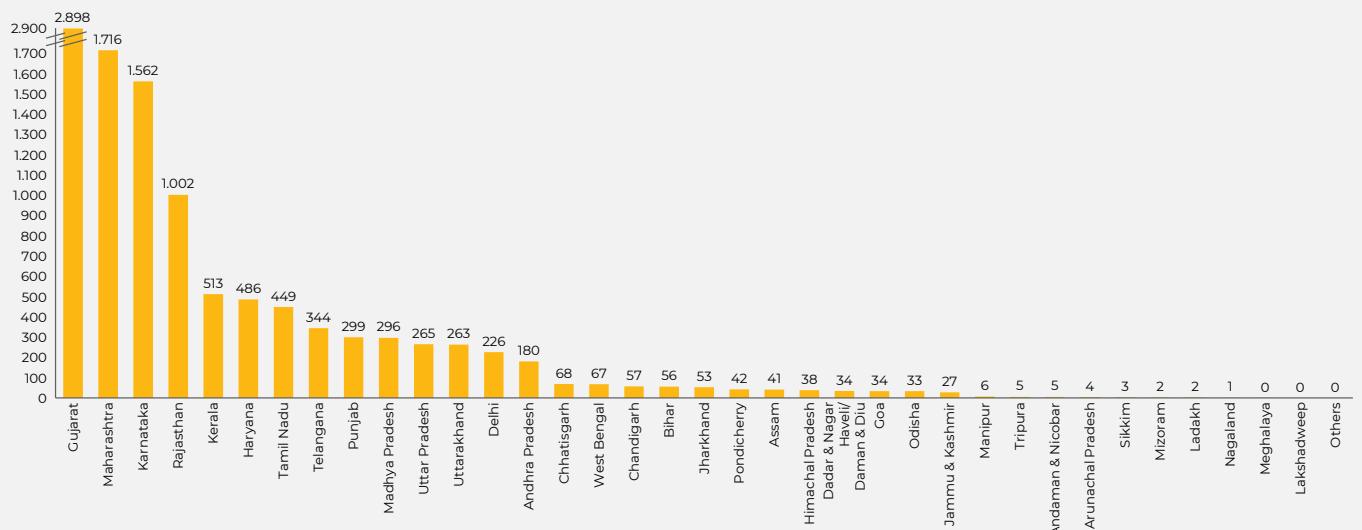
¹⁵ Ministry of New and Renewable Energy, Year- End Review 2022, 2022; Ministry of Power, Transmission Works under Green Energy Corridor – II, 2022

Figure 5: Solar power installed capacity by state and UT in MW (2023)



Source: Ministry of New and Renewable Energy, State-wise RE installed capacity, accessed November 2023

Figure 6: Rooftop solar power installed capacity by state and UT in MW (2023)



Source: Ministry of New and Renewable Energy, State-wise RE installed capacity, accessed on November 20

Solar power consumption

Ambitious policy targets and several policy and regulatory mechanisms for RE demand creation are expected to keep solar consumption on an upward trajectory.

The central government and DISCOMS are India's largest off-takers of utility-scale solar power:

The central government (through NTPC and SECI) and DISCOMS comprise 86% of utility-scale solar offtake. Commercial and industrial users make up the remainder.¹⁶ These are driven by various

¹⁶ Bridge to India, India RE Navigator, accessed June 2023

government measures like Declaration of trajectory for Renewable Purchase Obligation (RPO) up to year 2029-30.

Commercial and industrial (C&I) consumers comprise a growing share of solar demand:

In the rooftop solar segment, 70% of all installations by C&I consumers.¹⁷ The top five states comprising 50% of total C&I rooftop installations in solar energy are Rajasthan, Gujarat, Maharashtra, Tamil Nadu, and Karnataka.¹⁸ Residential demand has not grown significantly, with less than 1% of households in urban India installing rooftop solar systems.¹⁹

Various government schemes and programs aim to incentivise consumers to adopt solar energy:

Through the PM KUSUM scheme, farmers and other beneficiaries can sell excess power generated to DISCOMS, thus supplementing their income.²⁰ Other policy level interventions that facilitates solar energy purchase include a) Standard Bidding Guidelines for tariff based competitive bidding process for procurement of Power from Grid Connected Solar PV and Wind Projects, b) Launch of Green Term Ahead Market (GTAM) to facilitate sale of Renewable Energy Power through exchanges, c) Atal Jyoti Yojana, d) Preference to “Make in India” public procurement, e) Simplified procedure for RTS

Workforce development

India's solar energy sector plays a vital role in the country's renewable energy workforce, and several ongoing efforts focus on cultivating a skilled solar workforce.

India's solar sector employs a smaller portion of the renewable energy workforce than other large solar markets:

According to 2019-20 data, the solar energy sector in India accounted for 23% of the total renewable workforce.²¹ Within the solar workforce, rooftop solar is more labour-intensive than other renewables, providing 24.72 job-years per megawatt (MW) compared to 3.45 job-years per MW for ground-mounted solar. Solar photovoltaic module manufacturing creates employment of 2.60 job-years per MW of modules manufactured.²²

Ongoing efforts to build a solar workforce focus on training technicians:

The Suryamitra training program, initiated by the Ministry of New and Renewable Energy in 2015, trains youth in solar technology for installation, commissioning, and maintenance. So far, the program has certified over 78,000 trainees through government-supported schemes.²³ The Skill Council for Green Jobs, with support from MNRE, has been working to enlist credible training institutions nationwide to provide quality training in solar PV technology.²⁴

Achieving India's 2030 targets is expected to generate over 3 million jobs:

Experts estimate that meeting India's targets for installed solar capacity in 2030 will generate 2.7 million jobs in utility-scale solar and 0.5 million in rooftop solar.²⁵

¹⁷ Bridge to India, India RE Navigator- Rooftop solar, accessed June 2023

¹⁸ JMK Research and IEEFA, Powering Up Sunshine – Untapped Opportunities in India's Rooftop Solar Market, 2020

¹⁹ Bridge to India, India Residential Rooftop Solar Market 2023, 2023

²⁰ Ministry of New and Renewable Energy, Annual Report 2021-22, 2021

²¹ IRENA, Renewable Energy and Jobs Annual Review 2021, 2021

²² CEEW and NRDC, Greening India's Workforce, 2017

²³ Ministry of New and Renewable Energy, Suryamitra Programme, accessed June 2023

²⁴ Skill Council for Green Jobs, Solar PV, accessed June 2023

²⁵ CEEW, India's Expanding Clean Energy Workforce, 2022

Waste management

While the environmental impact of solar is relatively low compared to conventional sources, there are concerns about end-of-life management and disposal of panels and batteries.

End-of-life solar waste management is an impending concern given India's net-zero targets:

Although most installed systems in India are still new, the disposal of solar panels after their 25-year lifespan is an emerging issue. Several components used in PV panels and batteries are classified as potentially hazardous materials, and their improper disposal poses risks to human health and the environment.²⁶ The volume of waste from solar photovoltaic (PV) panels in India is projected to reach 200,000 tonnes by 2030 and approximately 1.8 million tonnes by 2050.²⁷ Researchers from the Indian Institute of Technology (IIT) Mandi have put forth industrial methods to recycle solar cell components and materials and how it can recover valuable resources like cadmium, tellurium, indium, gallium, and germanium, which are in limited supply and high demand within the industry.²⁸

New rules mandate proper management of solar PV module waste:

The Ministry of Environment, Forest and Climate Change issued the E-Waste (Management) Rules, 2022, which include regulations for handling solar PV modules. Manufacturers and producers of solar panels must register on the designated portal, store the generated waste according to guidelines set by the Central Pollution Control Board until 2034-2035, submit annual returns on the portal by the end of each year until 2034-2035, process non-solar waste under applicable rules, and maintain an inventory of solar modules on the portal.²⁹

Research & Development

While solar accounts for the largest share of R&D expenditure in renewable energy in India, funding levels remain low compared to other countries.

Need for increased R&D funding for solar energy in India:

Between 2015 and 2020, the Department of Science and Technology funded solar R&D projects worth USD 28 million (INR 208 crore). These funding efforts pale compared to other countries like South Korea, which allocated USD 60 million (INR 450 crore) specifically for solar manufacturing research in 2020. Similarly, the United States allocated USD 132 million (INR 990 crore) for solar PV R&D and manufacturing innovations in the same year.³⁰

There is an emerging startup ecosystem for solar energy:

Initiatives to provide backing for promising solar technologies that can cater to community concerns are gaining popularity. An example of such an effort is the BIRAC-Innovation Challenge Award, designed to empower Indian innovators in presenting renewable energy driven solutions for challenges related to clean cooking, such as solar powered cooking.³¹

International collaboration

The International Solar Alliance (ISA), launched by India and France in 2015, is an example of India's focus on advancing global solar energy cooperation. India also has strategic partnerships with technical institutions like the IEA and IRENA, and several strategic agreements and ongoing joint activities with countries like Russia and Vietnam.

²⁶ Bridge to India, Managing India's PV Module Waste, 2019

²⁷ Bridge to India, Managing India's PV Module Waste, 2019

²⁸ The Environment, IIT Mandi discovers ways to reduce environmental impact of solar cell waste, 2023

²⁹ Ministry of New and Renewable Energy, Solar Waste Treatment under E-Waste (Management) Rules 2022, 2023

³⁰ CEEW, Making India a Leader in Solar Manufacturing, 2022

³¹ BIRAC, SOCH-Innovation Challenge Award, 2021

Through ISA, India fosters global collaboration to accelerate grid interconnectivity, mobilise investment, and support deployment through capacity-building:

The One Sun One World One Grid initiative, proposed by India at the first ISA Assembly, seeks to interconnect regional grids and facilitate solar energy transmission globally. ISA trains bankers to assess solar PV projects' techno-commercial feasibility and financial viability.³² The Government of India supports ISA by training master trainers in solar energy through the Indian Technical and Economic Cooperation (ITEC) Scheme.³³ With support from the National Institute of Solar Energy, ISA launched a Fellowship Program for undergraduate students of member countries to provide training and capacity building in solar energy.³⁴

India partners with global agencies to boost efforts in advancing solar energy:

India has signed strategic partnerships with International Energy Agency (IEA) and International Renewable Energy Agency (IRENA) to promote renewable energy growth and clean energy transitions. In collaboration with CEEW and MNRE, IEA organised an online workshop on 'Unlocking the economic potential of rooftop solar energy in India' in 2020.³⁵

India is expanding its solar partnerships and activities with countries:

The Solar Energy Corporation of India (SECI) and the Russian Energy Agency (REA) signed a memorandum of understanding (MoU) to set up 500MW of solar photovoltaic (PV) projects in India between 2016 to 2022.³⁶ The National Solar Energy Federation of India (NSEFI) and the Vietnam Clean Energy Association (CEA) entered into a Memorandum of Understanding (MoU) in 2020, intending to foster the exchange of knowledge, best practices, and information between the two nations, and explore and pursue new business opportunities within the realm of solar energy.³⁷

Monitoring and evaluation

MNRE monitors large central schemes through appointed nodal officers, while state-level committees oversee the implementation of state-specific solar policies.

Monitoring of large central schemes comes under the purview of MNRE:

Typically, nodal officers are appointed to closely monitor and guide large central schemes. For example, under the scheme for the development of Solar Parks and Ultra Mega Solar Power Projects, the nodal officer monitors the scheme's progress and ensures the timely completion of all relevant activities.³⁸

At the state level, committees oversee the implementation of solar policies:

Committees are established to ensure the objectives and targets set by solar policies are achieved. For instance, the Tamil Nadu Energy Development Agency has established an inter-departmental monitoring and coordination committee to oversee the achievement of objectives and targets specified in the state solar policy. Similarly, the Solar Energy Policy of New Delhi mandates the formation of an apex committee to monitor policy implementation progress every quarter.³⁹

³² ISA, ISA's Solar Banking Initiative, accessed June 2023

³³ ISA, Solarizing Heating and Cooling Systems, accessed June 2023

³⁴ ISA, OSOWOG Initiative, accessed June 2023; Ministry of New and Renewable Energy, ISA Fellowship Program for Mid-Career Professionals, 2021

³⁵ IEA, Unlocking the Economic Potential of Rooftop Solar PV in India, 2021

³⁶ Ministry of New and Renewable Energy, Memorandum Of Understanding between Solar Energy Corporation of India and Russian Energy Agency regarding Construction of Solar Power Plants in Republic of India, 2016

³⁷ National Solar Energy Federation of India (NSEFI), MOUs, 2020

³⁸ Ministry of New and Renewable Energy, Implementation of a Scheme for Development of Solar Parks and Ultra Mega Solar Power Projects in the country commencing from 2014-15 and onwards, 2014

³⁹ TIDCO, Tamil Nadu Solar Energy Policy-2019, accessed June 2023; National Solar Energy Federation of India, Public Comments on draft Delhi Solar Policy 2022, 2023

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve by adopting effective solar practices. These revolve primarily around the potential for decentralised solar energy for off-grid applications, growth of green hydrogen and electric mobility sectors and increase in India's solar manufacturing capacity.



Accelerate India's energy security journey:

India has abundant solar energy resources, with approximately 5,000 trillion kilowatt-hours (kWh) of energy falling on its land area yearly, thus making solar the most secure of all energy sources.⁴⁰



Promote decentralised solar energy supply for off-grid applications:

The provision of decentralised solar power has the potential to meet the energy needs in remote and rural areas, alongside creating new jobs and promoting sustainable livelihoods in these regions. The government has already started taking efforts in this direction through the Off-grid and decentralised solar PV applications programme and PM KUSUM scheme, which support the deployment of distributed solar resources like solar streetlights and solar water pumps. Furthermore, MNRE's Promotion of DRE (Decentralized Renewable Energy) Livelihood Applications framework aims to create an enabling ecosystem for Decentralised Renewable Energy (DRE) adoption to boost sustainable livelihoods in India.⁴¹



Support the growth of other sectors like green hydrogen and electric mobility:

The growing focus on green hydrogen which will help India become a low-carbon economy, translates to the importance of renewables like solar energy used in its production. The decline in costs of technologies like solar PV and electrolyzers is considered one of the main driving factors for the achievement of cost-competitiveness by green hydrogen.⁴² Similarly, the powering of EVs by renewables will help India achieve green mobility in its true sense. Solar rooftop PV is an attractive technology for cost-effective and scalable deployment in India's EV sector.⁴³



Foster and continue to develop India's solar manufacturing capacity:

India ranks fourth in the world in solar power capacity. India must maintain this growth trajectory, and government efforts through Production Linked Incentive (PLI) schemes for solar PV modules display a strong commitment to developing an excellent solar manufacturing ecosystem in India.⁴⁴



Participate and lead in global solar energy innovation:

Advancements in manufacturing processes in recent years have increased solar efficiency in India, bringing down the levelised cost of electricity.⁴⁵ Further, India is already taking steps to capture the global stage in the solar energy system. Through the International Solar Alliance (ISA), it fosters global collaboration to accelerate grid interconnectivity, mobilise

⁴⁰ Ministry of New & Renewable Energy, Solar Energy, accessed July 2023

⁴¹ Ministry of New & Renewable Energy, Annual Report, 2021-22; Ministry of New & Renewable Energy, Framework for Promotion of Decentralised Renewable Energy Livelihood applications, 2022

⁴² The Energy and Resources Institute, The Potential Role of Hydrogen in India: A pathway for scaling-up low carbon hydrogen across the economy, 2020

⁴³ Center for Study of Science, Technology and Policy, Solar Energy-Based EV Charging: A Pilot and Techno-Economic Study, 2021

⁴⁴ Ministry of New and Renewable Energy, Year- End Review 2022, 2022

⁴⁵ CEEW, Making India a Leader in Solar Manufacturing, 2022

investment, and support deployment through capacity-building.⁴⁶ India must continue to engage global partners and emerge as a key contributor to drive and lead innovation in solar energy.



Contribute to the green jobs market:

The solar sector is expected to create 3.2 million jobs by 2030, including 2.7 million in rooftop solar and 0.5 million in utility-scale solar.⁴⁷

Challenges

This section highlights some of the major challenges and constraints facing solar energy in India. Efforts to resolve such challenges will ensure adequate capacity additions in solar energy and economical domestic manufacturing of quality solar modules.



India's current rate of capacity additions is insufficient to meet 2030 targets:

Although India added 13 GW of solar capacity in 2022, this is only half of what is required annually to reach the 2030 target of 280 GW.⁴⁸ India needs to add about 30 GW annually to meet its desired 2030 targets.



Domestically manufactured solar modules are 33% more expensive than those manufactured in China⁴⁹:

One of the biggest reasons for India's relatively more expensive PV modules is higher BOM (bill of materials) costs, i.e., costs of components required to build solar modules. These account for 56% of the price difference in modules compared to Chinese counterparts. China also has lower labour and overhead costs due to large-scale, automated manufacturing and lucrative financing options available to manufacturers. Additionally, India underutilises its existing manufacturing capacity, with capacity use at only 50%.⁵⁰



Quality and safety issues in solar PV systems:

The push for low prices in India's solar energy market often results in low-quality installation components and inferior system design and quality.⁵¹ According to a study conducted by the National Renewable Energy Laboratory (NREL), most concerns related to the quality and safety of solar photovoltaic (PV) systems arise either during the phase of procuring components, the installation stage and the design phase. Although Indian regulators have prescribed standards for solar PV projects via the Bureau of Indian Standards (BIS), these mainly focus on components and lack attention to workmanship-related concerns. Further, enforcing these standards is left to stakeholders, leading to lax implementation.⁵²



Land acquisition-related issues in utility-scale solar projects:

While NISE has estimated that 750 GW of solar PV would need only 3% of India's wasteland areas, the quantity of such wastelands has declined and may not always be ideal for solar projects. Furthermore, the process of getting approvals for agricultural land use for non-agricultural purposes like solar projects are lengthy. Additionally, some states restrict the maximum amount of land acquired through land ceiling limits.⁵³

⁴⁶ ISA, ISA's Solar Banking Initiative, accessed June 2023

⁴⁷ CEEW, India's Expanding Clean Energy Workforce, 2022

⁴⁸ Ministry of New and Renewable Energy, Year wise achievements, accessed December 2023; Ministry of New and Renewable Energy, Solar Energy capacity has nearly tripled in last 5 years from 21651 MW to 64380 MW, 2023

⁴⁹ CEEW, Scaling Up Solar Manufacturing to Enhance India's Energy Security, 2020

⁵⁰ CEEW, Scaling Up Solar Manufacturing to Enhance India's Energy Security, 2020; JMK Research and IEEFA, Viability Assessment of New Domestic Solar Module Manufacturing Units, 2021

⁵¹ USAID-NREL, Distributed Solar Quality and Safety in India, 2020

⁵² USAID-NREL, Distributed Solar Quality and Safety in India, 2020

⁵³ IEA, India Energy Outlook, 2021



Need for energy storage provisions:

CEA models project that India will require 34 GW of battery energy storage systems by 2030 to maintain grid stability.⁵⁴ However, existing energy frameworks largely exclude energy storage systems from key programs posing a barrier to investment in this space.⁵⁵ Presently, many states do not have a regulatory framework for grid scale battery storage. Moreover, mechanisms for integrating storage in the grid are limited. For instance, under the present net metering provisions, power exported from BESS (battery energy storage systems) to the grid is not permitted.⁵⁶



The growing interest of Commercial and Industrial (C&I) consumers is hampered by policy and regulatory challenges, amongst others:

Policy and regulatory uncertainty hampers RE procurement by the C&I segment as policies and regulations vary across states and may be inadequately implemented.⁵⁷ More specifically, the uptake of rooftop solar amongst MSMEs at the state level is hindered by issues such as inadequate financing options, low awareness about commercial and technical benefits, and policy uncertainty.⁵⁸

⁵⁴ Center for Science and Environment, Factsheet 6 Energy Storage, accessed June 2023

⁵⁵ National Renewable Energy Laboratory, U.S Department of Energy, Policy and Regulatory Readiness for Utility-Scale Energy Storage: India, accessed on December 2023

⁵⁶ IESA, India Stationary Energy Storage Market Overview Part 1: Front-of-the-Meter | FTM 2022 – 2030, 2023; IEA and NITI Aayog, Renewables Integration in India, 2021; JMK Research and IEEFA, Emerging Technology Trends in the C&I Rooftop Solar Market in India, 2021

⁵⁷ WWF, Renewable Energy Demand in India, 2019

⁵⁸ SUPRABHA and EY, Identifying barriers for rooftop solar uptake in MSMEs and development of a mitigating financial framework, 2020

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee solar energy.



NITI Aayog

Conducts policy analysis of energy sectors related to power, coal, petroleum and natural gas, new and renewable energy, and atomic power; and undertakes steps to improve energy planning, and data management in India



Ministry of New and Renewable Energy (MNRE)

Develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security



Bureau of Energy Efficiency

Assists in developing policies and strategies to promote energy efficiency and conservation and utilises existing resources and infrastructure in performing functions under the Energy Conservation Act



Central Electricity Authority

Acts as a technical support base for power sector stakeholders and supports policy formulation, establishing technical standards, project monitoring and human resource skilling



Central Electricity Regulatory Commission

Improves regional transmission system operations, formulates efficient tariff mechanisms, promotes power market development, and facilitates technological advancements alongside environmental and safety compliance



National Institute of Solar Energy

Works towards research and development activities, conducts testing and certification of solar components, carries out training programs, and fosters the growth of solar products and applications



Indian Renewable Energy Development Agency

Finances and supports self-sustainable investments in energy generation from renewable sources, and carry out promotion of energy efficiency and sustainable environmental technologies

Major policies and programs

This section highlights existing policies related to flood management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Jawaharlal Nehru National Solar Mission, 2010⁵⁹

The National Solar Mission aims to establish India as a global solar energy leader with the following key provisions:

- Promotes the use of solar energy in various sectors through various schemes, favourable policies for solar deployment and incentives
- Creates conducive conditions by scaling capacity and bringing about technological innovation to reduce costs



Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan (PM KUSUM), 2019⁶⁰

The PM KUSUM scheme aims to ensure energy security for farmers in India, along with fulfilling India's 2030 commitments of achieving 40% installed power capacity from non-fossil-fuel sources through the following key provisions:

- Component A provides for setting up 10 GW of grid-connected, decentralised RE power plants. These plants (500 kW to 2 MW capacity) can be set up by individual farmers, groups, and other specified entities, wherein DISCOMS would purchase generated power at pre-determined tariffs
- Component B provides for the installation of 17.5 lakh stand-alone solar agriculture pumps. Support will be provided to farmers for replacing existing diesel pumps with stand-alone solar water pumps (SWP) up to 7.5 HP in off-grid areas
- Component C provides for solarisation of 10 lakh grid-connected agriculture pumps. This will support individual farmers to solarise their pumps, and DISCOMS will purchase excess power at pre-fixed tariffs



National Wind Solar Hybrid Policy, 2018⁶¹

This policy provides a framework for the promotion of large grid-connected wind-solar PV hybrid systems for efficient utilisation of transmission infrastructure and land. It aims to reduce the variability in renewable power generation and achieve better grid stability.

⁵⁹ International Energy Agency, Jawaharlal Nehru National Solar Mission, 2021; Department of New and Renewable Energy, Government of Haryana, Introduction JNNSM, accessed July 2023

⁶⁰ National portal of India, Pradhan Mantri Kisan Urja Suraksha Evam Utthaan Mahabhiyan (PM KUSUM) scheme, accessed July 2023

⁶¹ Ministry of New and Renewable Energy, National Wind-Solar Hybrid Policy, 2018

National Mission for Enhanced Energy Efficiency, 2011⁶²

This mission consists of four specific energy efficiency initiatives under its umbrella that are aimed at facilitating energy-intensive industries (industries that consume large quantities of energy in the production process) to reduce their energy footprint and catalyse investments in the energy efficiency sector.

Green Energy Corridor Scheme⁶³

The Green Energy Corridor scheme aims to facilitate the evacuation and integration of renewable energy. It involves developing transmission systems and control infrastructure to increase India's overall RE capacity by harnessing the potential of RE-rich states like Andhra Pradesh, Gujarat, Himachal Pradesh, Jammu and Kashmir, Karnataka, Maharashtra, Rajasthan, Madhya Pradesh, and Tamil Nadu.

Rooftop Solar Programme⁶⁴

The initiative seeks to boost solar power production by encouraging greater use of rooftop solar installations. Presently, MNRE is executing the second phase of the Rooftop Solar Programme, with guidelines released in August 2019. This phase aims to add 4000 MW of rooftop solar (RTS) capacity, focusing on providing Central Financial Assistance (CFA) in the residential sector, including households in rural areas.

62 Bureau of Energy Efficiency, National Mission for Enhanced Energy Efficiency, accessed June 2023

63 Ministry of Power, Transmission Works under Green Energy Corridors-I, Accessed December 2023

64 Ministry of New and Renewable Energy, Rooftop Solar Programme Phase-II to provide subsidized rooftop solar power connections to rural population, 2022

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. Some ideas to enable progress given India's ambitious solar energy goals include:

 Increase self-reliance and promote domestic equipment production	Increase investment in research and development to foster innovation and develop cutting-edge technologies domestically - Incentivize and support research partnerships with global actors, including solar R&D companies - Increase government-led R&D expenditure into new solar technologies and decentralised applications that are not dependent on existing supply chains	Support the development of a skilled workforce across the solar value chain - Foster ITI collaboration with industry to design and deliver skilling courses on solar component manufacturing and the operations & maintenance of solar plants with support from the Skills Council for Green Jobs, the National Apprenticeship Scheme, and the Skill India Mission - Enable industry-academia partnerships through collaboration platforms that convene key stakeholders to discuss opportunities and challenges in existing skilling curricula
 Further enhance solar transmission and reduce curtailment losses	Expand effective demand response management - Expand time-of-use tariffs to cover all sectors, not just industrial and commercial ones, with peak-hours surcharges and off-peak-hours rebates - Mandate State Load Despatch Centres to publicly make their load curve data available -	Strengthen state transmission and storage infrastructure and networks - Upgrade existing grid infrastructure to make it more robust and flexible by investing in smart grid technologies that allow for better management and distribution of solar power (including through support via green bonds and other concessional finance to develop infrastructure as part of the Green Energy Corridor) - This could also include expanding the transmission network to connect more remote solar power installations to the main grid can help reduce transmission losses - Increasing adoption of advanced energy storage solutions, especially grid-scale - Enhancing forecasting and scheduling of power by encouraging the development and implementation of more sophisticated forecasting tools that use data analytics, AI, and machine learning to more accurately predict solar power generation



**Innovative
mechanisms
to drive
adoption of
rooftop solar**

**Tailored demand-side interventions
among different user segments**

- Stimulate financing by offering special-purpose loans and guarantees to consumers with low creditworthiness

Continue enhancing consumer awareness through targeted information, education, and communication campaigns

**Explore DISCOM-led business model
along with financial incentives**

- Explore and execute innovative DISCOM-led business models, such as on-bill financing and solar partner models
- Offer additional financial incentives like performance-based incentives to DISCOMs
- Enhance administrative efficiency of RTS installation by providing technical assistance to DISCOMs and encouraging DISCOMs to proactively use the national portal for rooftop solar

Recommended readings

CEEW. (2023). *Community Solar for Advancing Power Sector Reforms and the Net zero Goals*
<https://www.ceew.in/publications/community-solar-advancing-power-sector-reforms-and-net-zero-goals>

CEEW. (2022). *India's Expanding Clean Energy Workforce*
<https://www.ceew.in/sites/default/files/Green-Jobs-Report-Jan27.pdf>

NITI Aayog and IEA. (2021). *Renewables Integration in India*
<https://iea.blob.core.windows.net/assets/7b6bf9e6-4d69-466c-8069-bdd26b3e9ed1/RenewablesIntegrationinIndia2021.pdf>

IEA. (2021). *India Energy Outlook*
https://iea.blob.core.windows.net/assets/1de6d91e-e23f-4e02-b1fb-51fdd6283b22/India_Energy_Outlook_2021.pdf



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